

1. **Standard Geometric Series**

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}, \quad |r| < 1$$

2. **Geometric Series with First Term a**

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, \quad |r| < 1$$

3. **Alternating Geometric Series**

$$\sum_{k=0}^{\infty} (-1)^k r^k = \frac{1}{1+r}, \quad |r| < 1$$

4. **Series Starting at $k = 1$**

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r}, \quad |r| < 1$$

5. **Weighted Geometric Series (Linear)**

$$\sum_{k=1}^{\infty} kr^k = \frac{r}{(1-r)^2}, \quad |r| < 1$$

6. **Weighted Geometric Series (Quadratic)**

$$\sum_{k=1}^{\infty} k^2 r^k = \frac{r(1+r)}{(1-r)^3}, \quad |r| < 1$$

7. **Weighted Geometric Series (Cubic)**

$$\sum_{k=1}^{\infty} k^3 r^k = \frac{r(1+4r+r^2)}{(1-r)^4}, \quad |r| < 1$$